14.3 Memory allocation of objects

The member functions are created and placed in the memory space only when they are defined as a part of the class specification. Since all the objects belonging to that class use the same member function, **no separate space is allocated for member functions when the objects are created. Memory space required for the member variables are only allocated separately for each object** because the member variables will hold different data values for different objects

Illustration 14.3 Memory allocation for objects

```
# include <iostream>
using namespace std;

class product
{
        int code, quantity;
        float price;
        public:
        void assignData();
        void Print();
};
int main()
{
        product p1, p2;
        cout<<"\n Memory allocation for object p1 " <<sizeof(p1);
        cout<<"\n Memory allocation for object p2 " <<sizeof(p2);
        return 0;
}
```

Output:

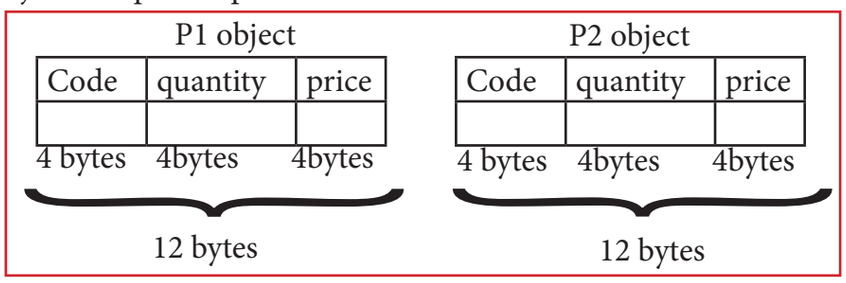
```
Memory allocation for object p1  12
Memory allocation for object p2  12
```

Member functions assignData( ) and Print( ) belong to the common pool in the sense both the objects p1 and p2 will have access to the code area of the common pool.

Note

The members will be allocated with memory space only after the creation of the class type object

Memory for Objects for p1 and p2 is illustrated:



Chapter 14 Page 233-248.indd 238 3/24/2020 9:23:04 AM